

EV Charging Infrastructure Intelligence Dashboard

Complete Project Documentation, Insights and Conclusion

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Project Type: Data Analytics / Business Intelligence

Tool: Microsoft Power BI

Primary Technologies: Power Query, DAX, Power BI Visualizations, Maps, Drillthrough

The project specification defines four main phases: data cleaning and preprocessing, calculated columns, DAX measures, and a three-page interactive dashboard.

1. Project Overview

The EV Charging Infrastructure Intelligence Dashboard is a Power BI-based data analytics project designed to analyze electric vehicle charging station infrastructure. The project takes raw EV charging station data and transforms it into an interactive business intelligence dashboard. The dashboard helps users understand:

- Where EV charging stations are located
- Which countries have the most stations
- Which charging operators manage the most stations

- Which connector types are commonly available
- How many stations are operational
- How charging stations are distributed by capacity
- How charging infrastructure differs between countries and regions
- Details of individual charging stations

The final dashboard contains three major analytical pages:

1. Executive Overview
2. Infrastructure Analysis
3. Station Details & Drillthrough Analysis

The uploaded dashboard shows the completed three-page Power BI report with KPIs, maps, charts, tables, capacity analysis, regional analysis and station-level information.

2. Problem Statement

The growth of electric vehicles requires a reliable and well-distributed charging infrastructure. However, raw EV charging station datasets are difficult to analyze directly because they may contain:

- Missing city and state information
- Different country codes
- Inconsistent operator names
- Multiple connector types stored in one field
- Different station status values
- Geographic coordinates that need validation
- Duplicate station records
- Different numbers of connectors per station

Without proper cleaning and transformation, it becomes difficult to answer important questions such as: Which country has the highest number of charging stations? Which operator has the largest charging network? What percentage of stations are operational? Which connector types are most common? Which regions have the strongest charging infrastructure?

The purpose of this project is therefore to transform raw EV station data into a reliable and interactive analytical dashboard.

3. Project Objectives

Main Objective

To develop an interactive Power BI dashboard that analyzes EV charging infrastructure using data cleaning, transformation, DAX calculations, geographic analysis and interactive visualizations.

Specific Objectives

- Clean and preprocess the raw EV charging station dataset.
- Handle missing and inconsistent values.
- Standardize country names, charging operator names, and station status values.
- Split multiple connector types into individual values.
- Validate latitude and longitude, and remove duplicate stations.
- Create charging capacity categories, calculated columns, and analytical measures using DAX.
- Build interactive Power BI dashboard pages providing country, operator, connector-level, and drillthrough analysis.
- Generate useful business insights from the data.

4. Dataset Description

The uploaded CSV dataset contains 10,000 EV charging station records and 14 original columns. The dataset contains stations from multiple countries and regions and supports geographic, operator, connector and operational analysis.

Column	Description
id	Unique station identifier
title	Charging station name
address	Physical station address
town	City or town
state	State or province
postcode	Postal code
country	Country code
lat	Latitude
lon	Longitude
operator	Charging station operator
status	Current station status
num_connectors	Number of connectors
connector_types	Available connector types
date_added	Date and time when station was added

5. Data Quality Analysis

Before building the dashboard, the dataset was examined for data quality problems.

Missing Values

Column	Missing Records
Address	1
Town/City	203
State	2,869
Postcode	1,670
Operator	598
Connector Types	5

Handling Strategy:

- **State/Postcode/Address:** Retained to avoid unnecessarily reducing the dataset.
- **City/Town:** Missing values handled during preprocessing.
- **Operator:** Standardized as *"Unknown Operator"* to prevent blank categories.
- **Connector Types:** Handled as unknown where necessary.

6. Data Cleaning and Preprocessing

Step 1: Handle Missing Values

Missing values were managed intelligently to avoid losing useful station records while ensuring dashboard visual integrity.

Step 2: Remove Duplicates

Duplicate station records were checked. The dataset contained no exact duplicate rows in the original CSV analysis, thus all 10,000 original records were retained.

7. Rename Columns

For easier use in Power BI, technical column names were transformed into readable business names.

Original	Dashboard Name
id	Station ID
title	Station Name
town	City/Town
country	Country Code
operator	Charging Operator
num_connectors	Number of Connectors
connector_types	Connector Types
date_added	Date Added

8. Trim and Clean Text

Text columns (Country, Operator, Connector Types, City, State, Address) were cleaned using Power Query to remove unnecessary spaces (e.g., " Chargepoint " → "Chargepoint").

9. Standardize Country Names & Operator Names

Country codes (CA, US, ES, RU, etc.) were mapped to readable names (Canada, United States, Spain, Russia). Operator names were standardized (e.g., chargepoint → ChargePoint, flo → FLO) to prevent data fragmentation. Circuit Electrique emerged as the leading operator post-cleanup.

10. Standardize Status & Split Connector Types

Six original status values were streamlined (e.g., "Partly Operational (Mixed)" → "Operational"). Connector types packed into single fields (e.g., CCS (Type 2) | CHAdeMO) were split by delimiters to allow independent tracking of Type 1, Type 2, CCS, etc.

11. Validate Geography & Categorize Capacity

Latitude (-90 to +90) and Longitude (-180 to +180) were validated. Stations were grouped by capacity:

- **Small (1 connector):** ~7.1K stations (70.99%)
- **Medium (2-4 connectors):** ~2.8K stations (27.89%)
- **Large (5+ connectors):** ~0.1K stations (~1.1%)

12. Phase 2: Calculated Columns

Calculated columns were built using DAX to enrich the dataset:

```
Station Age = DATEDIFF(ev_stations_2025[Date Added], TODAY(), YEAR)
```

```
Connector Category = SWITCH(TRUE(),
    ev_stations_2025[Number of Connectors] = 1, "Single Connector",
    ev_stations_2025[Number of Connectors] <= 4, "Multi Connector",
    ev_stations_2025[Number of Connectors] >= 5, "High Capacity", "Unknown")
```

```
Operational Flag = IF(ev_stations_2025[Status] = "Operational", "Yes", "No")
```

13. Phase 3: DAX Measures

DAX measures calculate the core KPIs of the dashboard.

```
Total Stations = COUNTROWS(ev_stations_2025)
Operational Stations = CALCULATE(COUNTROWS(ev_stations_2025),
    ev_stations_2025[Status] = "Operational")
Non Operational Stations = CALCULATE(COUNTROWS(ev_stations_2025),
    ev_stations_2025[Status] <> "Operational")
Total Connectors = SUM(ev_stations_2025[Number of Connectors])
Average Connectors per Station = AVERAGE(ev_stations_2025[Number of Connectors])
Operational % = DIVIDE([Operational Stations], [Total Stations], 0)
```

14. Dashboard Page 1: Executive Overview

Provides a macro view of the infrastructure. KPI Results display ~10K Total Stations, ~9K Operational Stations, 626 Non Operational (93.46% Operational rate), and ~14K Total Connectors. Canada (3.8K) and the US (2.2K) dominate the dataset.

Insight: The 93.46% operational rate indicates a highly reliable grid in the sampled data, though the remaining 6.54% non-operational stations (626 units) warrant targeted maintenance due to the sheer physical volume they represent.

15. Dashboard Page 2: Infrastructure Analysis

Provides deeper infrastructure breakdowns. Circuit Electrique leads operators with 2.0K stations, followed by ChargePoint (1.3K) and FLO (0.7K). The average of 1.46 connectors per station confirms the heavy dominance of "Small" capacity charging points.

Region	Total Stations	Operational	Non Operational
North America	6,092	5,828	264
Europe	2,017	1,792	225
Asia	782	701	81
Africa	330	330	0

16. Dashboard Page 3: Station Details & Drillthrough

Utilizes Drillthrough mechanics allowing users to right-click summary visuals (e.g., 'Canada' or 'ChargePoint') and drill down to a granular matrix table showing exact Station IDs, Locations, Connector Types, and Statuses, alongside zoomed-in map visuals.

17. Major Project Insights & Findings

- **High Operational Availability:** With 93.46% uptime, the recorded network demonstrates strong reliability.
- **North American Data Bias:** Out of 10K records, over 6K are located in North America (primarily Canada and the US), indicating the dataset leans heavily on western charging behaviors.
- **Small Stations Dominate:** 71% of stations possess only 1 connector. The global paradigm represented here is decentralized micro-locations rather than massive charging hubs.
- **Metadata Quality Needs Focus:** With roughly 400 stations falling under "Unknown Operator" and a massive data ingest spike logged in June (9.4K records), data sourcing methodologies could be refined.

18. Limitations & Future Scope

Limitations: The data heavily reflects a localized timeframe and geography (North America). Connector arrays often require careful counting to avoid duplications. Furthermore, without kWh output data, it's impossible to distinguish Fast DC chargers from Level 2 AC chargers.

Future Scope:

- Live API integration for real-time station availability tracking.
- Inclusion of Charging Power (kW) metrics to analyze rapid charging coverage.
- Integration of EV registration density maps to measure regional demand vs. supply.
- Predictive machine learning models to forecast station breakdowns and optimal expansion zones.

19. Final Conclusion

The EV Charging Infrastructure Intelligence Dashboard successfully transforms a raw dataset into an interactive analytical tool. By executing diligent Data Cleaning (handling missing values, standardizing syntax) in Power Query and formulating robust logic (DAX), the project uncovers that the modeled infrastructure is highly operational but primarily composed of small, single-connector units heavily clustered in North America. This Power BI dashboard is a functional blueprint for network operators and city planners striving to make data-driven, strategic EV grid expansions.